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Forget Miltank:
A Quantitative Search for the Best Team in Pokémon HeartGold and SoulSilver

A simulation study of 34 story battles, 219 obtainable Pokémon and every repeatable source of experience in Johto and Kanto

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Abstract

Pokémon HeartGold and SoulSilver are, underneath, entirely quantitative games: every battle is resolved by fixed formulas for stats, damage, experience and catch rates. This study uses those formulas to answer a question every player has asked: what is the best team to beat the whole game, where do you catch it, what should it know and hold, and how do you train it fastest? All data are read from the fan decompilation of the game's code, covering all 738 trainers, every wild encounter table and the location of every TM and held item, and battles are simulated with Pokémon Showdown's Generation 4 engine. Only Pokémon obtainable in a single cartridge, without trading, legendaries or events, are allowed. After an analytic screen of about 110 evolutionary lines and full simulations of the most promising teams, the 15 finalists are played through the 34 story battles in order, with only the moves, single-use TMs and held items they could really have at each point, and a level that never goes down. A core of Graveler, Heracross, Umbreon and Lapras appears in almost every good team. The best team, Feraligatr, Arcanine, Graveler, Heracross, Lapras and Umbreon, is complete before Chuck and never needs to be above an opponent's level until the Elite Four, which it must enter at level 54, four levels above Lance's strongest Pokémon; without held items it would need level 57. It beats Whitney one level below her Miltank, which is hard mainly for a starter fighting alone. Red requires a separate team built around Blissey, Tyranitar, Steelix and Tentacruel, trained to level 84, and the fastest way to get there is the Elite Four rematch, which gives about 75% more experience per minute than Mt. Silver. The complete plan, with every member's moves and held item in every battle, is given in an appendix.

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1. Introduction

Every Pokémon player remembers the moment the game stopped being easy. For a whole generation of players it was Goldenrod City's gym, where Whitney's Miltank rolled over team after team with Rollout, healed itself with Milk Drink and, for good measure, infatuated anything of the opposite gender with Attract. For others it was Clair's Kingdra, the Elite Four, or the long climb up Mt. Silver to face Red, whose Pikachu sits at level 88 at the top of the mountain. The usual answer to these walls is the same: go back to the tall grass and train.

Behind this folklore sits a game that is, underneath, entirely quantitative. Every Pokémon has six base stats, a type, a level and a list of moves; every attack is resolved by a published damage formula; experience, catch rates and even the odds of a wild encounter are fixed numbers stored in the game's code. This makes Pokémon HeartGold and SoulSilver (Game Freak, 2009), the 2009 remakes of the second-generation games and the author's favourites, a good test case for a simple question: if you do the maths properly, what is the best team to beat the whole game, where do you catch it, and what is the fastest way to train it?

This paper answers that question with three rules that keep it honest. First, the team must be made of Pokémon a player can actually obtain in a single cartridge of HeartGold or SoulSilver, with no trading, no legendary Pokémon and no event distributions. Second, every number comes from the game itself: trainer teams, wild encounter tables, evolutions, movesets and item locations are read directly from the fan decompilation of the game's code (pret, n.d.), and battles are played out with Pokémon Showdown's Generation 4 battle engine (Smogon, n.d.-a). Third, a team is judged by what matters to a player: playing the game's 34 story battles in order, from the first rival fight to Blue in Viridian City, with only the moves, TMs and held items it could really have at each point, how far above its opponents' levels must it be to win each battle at least nine times out of ten? A separate section then does the same for Red.

The answers, in order, are these. The best team found is Feraligatr, Arcanine, Graveler, Heracross, Lapras and Umbreon, available in HeartGold with Totodile as the starter; in SoulSilver, where Arcanine cannot be caught, Crobat takes its place. The team is assembled in that order of usefulness almost by accident: Geodude is caught in the first minutes of the game, Growlithe before Bugsy, Heracross from a Headbutt tree before Whitney, Eevee from Bill before Morty and Lapras in Union Cave before Chuck. Played through battle by battle, with the moves it knows and the items it has found at each point, this team never needs to be above an opponent's level until the Elite Four, which it must enter at level 54, four levels above Lance's strongest Pokémon; it beats Whitney at level 18, one level below her Miltank, and Blue at 57, three below his. Held items are worth three levels at the Elite Four, where the team would otherwise need level 57. Whitney's Miltank deserves its reputation only for a player who relies on the starter: on its own, each starter needs 8 to 11 extra levels to beat her. Red is a different problem. The best team of catchable Pokémon for that fight is built around Blissey, Tyranitar, Steelix and Tentacruel, and it needs level 84 to win nine times out of ten. Finally, experience in

these games does not depend on the winner's level, so the fastest training is always against the highest-level opponents available: Victory Road before the Elite Four, and the Elite Four rematches once all sixteen badges are won, which give about 75% more experience per minute than Mt. Silver.

The rest of the paper is short. Chapter 2 explains the handful of formulas and rules that run the game. Chapter 3 describes how the data were extracted and the battles simulated. Chapter 4 presents the results: what can be caught and when, the best team for each starter, how to assemble it and what it should know and hold in each battle, where the real difficulty walls are, the team that beats Red, and how to train it. Chapter 5 turns the results into a practical guide and lists what the model leaves out, and Chapter 6 concludes. Appendix C gives the complete plan, battle by battle.

2. How the Game Works

Four formulas do almost all the work in a Pokémon game: the ones that turn base stats into actual stats, attacks into damage, defeated Pokémon into experience, and thrown balls into caught Pokémon. A few rules about moves and held items complete the picture. The versions below are those of the fourth generation, which HeartGold and SoulSilver use, and each has been checked against the game's own code.

2.1 Stats and levels

Each species has six base stats: HP, Attack, Defense, Special Attack, Special Defense and Speed. An individual Pokémon adds individual values (IVs, from 0 to 31) and effort values (EVs, earned by battling), and its level scales the result:

$$\text{Stat} = \lfloor (2B + IV + \lfloor EV/4 \rfloor) \times L / 100 \rfloor + 5$$

where B is the base stat, L the level and IV and EV the individual and effort values. HP uses the same formula with L + 10 added instead of 5, and a Pokémon's nature can raise one stat by 10% and lower another by 10%. Two consequences matter here. Level enters every stat almost linearly, so a Pokémon ten levels higher is roughly 20% stronger at level 50, and base stats matter enormously: at the same level, a Pokémon with base Attack 130 hits about twice as hard as one with base Attack 65.

2.2 Damage

When a Pokémon uses an attacking move, the damage is

$$\text{Damage} = [(2L/5 + 2) \times Power \times A/D / 50 + 2] \times STAB \times Type \times R$$

where Power is the move's base power, A and D are the attacker's attacking stat and the defender's defending stat (Attack against Defense for physical moves, Special Attack against Special Defense for special moves), STAB is 1.5 when the move shares a type with its user, Type is the type-effectiveness multiplier (0, 0.25, 0.5, 1, 2 or 4), R is a random factor between 0.85 and 1, and critical hits, which happen one time in sixteen, double the damage. The formula explains why type matchups dominate the game: a super-effective move with STAB does three times the damage of a neutral move of the same power, which is more than twenty levels of difference would give.

2.3 Experience

Experience depends only on the Pokémon that was defeated, not on the level of the Pokémon that defeated it. In the game's code (pret, n.d.), a defeated Pokémon gives

$$\text{Experience} = b \times L / 7$$

where b is its species' base experience yield and L its level, divided equally between all the Pokémon that took part in the battle. The result is multiplied by 1.5 in trainer battles, by 1.5

again if the Pokémon holds a Lucky Egg, and by 1.5 if it was obtained in a trade. Because there is no scaling for the winner's level, as there is in later games, training a level-80 Pokémon on level-40 opponents is just as efficient as training a level-40 one. What matters is only how many high-level, high-yield Pokémon can be defeated per minute. Each species then needs a fixed total of experience to reach each level, following one of six growth curves; a Pokémon in the "slow" group needs 1,250,000 experience points to reach level 100, one in the "fast" group needs 800,000.

2.4 Catching

When a ball is thrown, the game first computes a modified catch rate

$$a = (3H_{max} - 2H) \times rate \times Ball / (3H_{max}) \times Status$$

where H_{max} and H are the target's maximum and current HP, rate is its species' catch rate (from 3 for legendaries to 255 for the commonest Pokémon), Ball is 1 for a Poké Ball, 1.5 for a Great Ball and 2 for an Ultra Ball, and Status is 2 if the target is asleep or frozen and 1.5 if it is paralysed, poisoned or burned. If a is 255 or more the Pokémon is caught; otherwise the ball "shakes" four times and each shake succeeds with a probability that rises with a. Taking the target down to low HP triples a; putting it to sleep doubles it again.

2.5 Moves and held items

A Pokémon knows at most four moves. It learns new ones by levelling up, each at a fixed level, and a move that is not learned at that moment, or is later replaced, is lost until the player reaches the Move Relearner in Blackthorn City, who teaches back any level-up move of the Pokémon's current form in exchange for a Heart Scale. A Pokémon that evolves keeps the moves it knows but stops learning those of its earlier form, so a move of the earlier form that comes after the evolution level can never be learned. TMs teach a move to any compatible Pokémon, but in this generation each TM disappears once used; only those sold in the Goldenrod and Celadon department stores can be bought again. HMs can be used any number of times, but an HM move can only be forgotten with the help of the Move Deleter, also in Blackthorn City.

Each Pokémon can also hold one item. The useful ones for a playthrough are simple: type-boosting items such as Mystic Water or Charcoal raise the power of moves of one type by 20%; a Sitrus Berry restores a quarter of the holder's HP once, when its HP falls below half; Leftovers restore a sixteenth of its HP every turn. Items can be moved between Pokémon at any time outside battle.

3. What I Did

This chapter explains how the game data were extracted, what counts as an obtainable Pokémon, how battles were simulated and how the best team was searched for. Every step is automated and reproducible from the code provided with this paper.

3.1 Reading the game's data

All game data come from pokeheartgold, a community project that has reconstructed the source code and data files of HeartGold and SoulSilver from the original cartridge (pret, n.d.). From its files I extracted:

• the base stats, types, abilities, catch rates, experience yields, growth rates and evolution rules of all 493 species;

• the level-up moves of every species, and which TMs and HMs each can learn;

• all 738 trainers with their exact teams, levels, moves, held items and the healing items they use in battle;

• the wild encounter tables of every route and cave, separately for HeartGold and SoulSilver and for morning, day and night, together with the Headbutt trees;

• the location of every TM and HM, read from the game's scripts (gifts and gym rewards), from the item balls placed on each map and from the stock lists of the Goldenrod and Celadon department stores;

• the location of every useful held item, read the same way and checked by hand against the official guides (The Pokémon Company International, 2010a, 2010b) and a community reference (Bulbapedia, n.d.), in particular for the weekly gifts of the seven siblings, such as Arthur's Hard Stone on Route 36 on Thursdays.

The extracted data match the familiar facts of the game, as listed in the official guides (The Pokémon Company International, 2010a, 2010b). Whitney's Miltank is level 19 and knows Rollout, Attract, Stomp and Milk Drink; Red's team is Pikachu (level 88), Lapras (80), Snorlax (82) and Venusaur, Charizard and Blastoise (84), with four Full Restores in his bag.

3.2 What counts as available

A Pokémon counts as available at a given point of the story if a player with a single cartridge can have it in the party at that moment. The story is divided into 35 checkpoints, one for each boss battle in the order in which a player meets them: the six battles against the rival, the eight Johto gym leaders, the Team Rocket executives, the Elite Four and Lance, the eight Kanto gym leaders, Blue and finally Red. Each wild area was assigned the checkpoint from which it can first be reached, taking into account the HMs and story events that open it (for example, Surf only after Morty, Route 47 only after the SecretPotion is delivered to Jasmine (Bulbapedia, n.d.), Mt. Silver only with all sixteen badges). Fishing requires the relevant rod and Headbutt trees the Headbutt tutor in Ilex Forest.

On top of the wild Pokémon, the gifts, static encounters and in-game trades that need no second cartridge were added by hand: the three starters, the Togepi egg, Bill's Eevee, the Shuckle in Cianwood, the Friday Lapras in Union Cave, the red Gyarados, the Rocket Hideout Electrode, the Dratini in the Dragon's Den and the Snorlax in Vermilion. Evolutions were then applied: by level, by friendship, by knowing a move, or by an evolution stone once it can be bought. Before the Johto league, Fire, Water, Thunder, Leaf and Moon Stones can be bought at the Pokéathlon Dome on set weekdays and a Sun Stone won in the Bug-Catching Contest, while Shiny, Dusk and Dawn Stones only appear after the National Pokédex (Altissimo, n.d.). Evolution by trade, legendary Pokémon, event Pokémon, the Pokéwalker, the Safari Zone's object-dependent encounters, radio-dependent encounters and swarms were all excluded.

The same rule applies to moves and items: a move or an item counts only from the point of the story at which it can be obtained. Of the 100 TMs and HMs, 84 can be obtained this way; the Game Corner's prize TMs are excluded because they require buying coins. For held items, the Life Orb is excluded because the Ruins of Alph chamber that holds it only opens with Ho-Oh in the party, and the Battle Frontier's items (Choice Band, Choice Scarf and others, bought with Battle Points) only become available after the story, for the fight against Red. How moves, TMs and items are then assigned to the team is described in Section 3.4.

3.3 Simulating battles

Battles were simulated with the Generation 4 battle engine of Pokémon Showdown (Smogon, n.d.-a), a widely used open-source simulator that reproduces the mechanics of HeartGold and SoulSilver, including abilities, held items, status conditions, critical hits and every move effect. Expected damage for decision-making was computed with the Showdown damage calculator (Smogon, n.d.-b). Both sides follow simple, stated rules.

The player picks, each turn, the move with the highest expected damage, adjusted for accuracy and for moves that need a charging or recharging turn. When a Pokémon faints, the player sends in the team member with the best matchup, and switches voluntarily only out of a clearly lost matchup to a teammate that takes less than 40% from the opponent's best attack. The player uses no items such as Potions during battle; its Pokémon hold the items the plan gives them (Section 3.4).

The trainer follows a simplified version of the game's own battle AI: it uses the move with the highest expected damage 80% of the time and a random move from its moveset 20% of the time, which lets status moves such as Attract, Thunder Wave or Sleep Powder appear at a realistic rate. It sends in its Pokémon in order and uses its healing items, exactly as stored in the game data, whenever its active Pokémon falls below a quarter of its HP. Trainer Pokémon have the individual values the game gives them; the player's Pokémon have average individual values (15) and no effort values, which makes the results slightly conservative.

3.4 Measuring a team, and searching for the best one

Testing every possible team is impossible: with about 110 candidate evolutionary lines there are over a hundred million ways to complete a team of six around a starter. The search therefore has three stages, each more realistic and more expensive than the last.

The first is a fast analytic screen: for every candidate line, every boss and three levels (the boss's level and four and eight levels below), a one-on-one duel with each of the boss's Pokémon is scored by the number of turns each side needs to knock the other out, using the damage formula. A team's score is the share of boss Pokémon that at least one member beats, summed over all bosses and levels, and a local search with thirty random restarts finds the best-scoring teams for each starter. The second stage runs the full battle simulation on the shortlisted teams. For each of the 34 story battles it finds the required level, the lowest level, applied to every team member, at which the team wins at least 90% of 40 simulated battles, with moves chosen afresh for each battle from everything the team could know by then. This stage selected the 15 finalists, the best two or three teams for each starter and version.

The third stage, which produces every result in Chapter 4, plays each finalist through the story as a player would, battle after battle, keeping track of what each Pokémon actually knows and holds. A Pokémon joins the team at the first battle for which it is available, knowing the moves it has when caught or received, and evolves as soon as it can. Before each battle, each member picks its four moves from the moves it already knows, the level-up moves it has reached since the previous battle, the HMs and department-store TMs available, and, from Clair onwards, anything the Move Relearner can teach; a move it drops is lost unless it can be relearned. Moves are chosen to maximise expected damage against the coming trainer's team, with half weight on the next three bosses so that a useful move is not thrown away. Each single-use TM goes, once, to the member that gains most from it over the next few battles, and only if the gain is substantial. Held items are then handed out before each battle, one copy to one Pokémon, to the members that gain most from them against that trainer, and are swapped freely between battles. The five battles against the Elite Four and Lance are fought in a row, as in the game, so the team faces them at one level with one set of moves chosen for all five, changing only held items between them.

The team's level can only go up. For each battle, the team keeps its current level if it wins at least 90% of the time there, and otherwise is levelled up to the lowest level that does. Teams are compared by the training they force beyond the bosses' own levels: over the battles for which the team had to be levelled up, the number of levels by which it then exceeds the boss's highest level. A team that is above a boss's level only because it trained for an earlier, harder battle is not penalised for it. Ties are broken by the average difference between the team's level and the boss's level. The ranking uses 40 battles per level; the best team of each starter was then played through again with 400 battles per level, which gives the levels and win rates reported for it, and once more without held items to measure what the items are worth.

The same stages were used for the endgame: the screen was run against Red and the Elite Four and Lance rematch teams with every obtainable Pokémon allowed, and the shortlisted teams were simulated against Red between levels 50 and 100, with held items, including the Battle Frontier's, and without the single-use TMs the story team has already spent.

3.5 Training and catching

The experience per minute of every repeatable source was computed from the encounter tables and trainer data with the formula of Section 2.3, under stated time assumptions: a wild encounter, including the search and a one-hit knockout, takes 30 seconds; a trainer battle takes one minute plus 20 seconds per Pokémon; and a full Elite Four and Lance run adds four minutes of walking and healing. Changing these assumptions scales the numbers but hardly changes the ranking. Catch probabilities follow the formula of Section 2.4 using the game's exact integer arithmetic.

4. What I Found

This chapter presents the results in the order a player would need them: what can be caught and when, the best team for the story, how to assemble it and what each member should know and hold in each battle, where the story's real difficulty lies, the team that beats Red, and how to train both teams.

4.1 What you can catch, and when

A single cartridge gives access to 219 different species without trading, legendaries or events, in both HeartGold and SoulSilver. The two versions share 204 of them and each has 15 exclusives, such as Growlithe, Mankey and Gligar in HeartGold and Vulpix, Meowth and Skarmory in SoulSilver. Figure 1 shows how quickly the choice widens. Only 29 species are available before the first rival battle and 41 before Falkner, but the number jumps to 101 before Whitney, once the Headbutt trees, the Pokéathlon evolution stones and Goldenrod's surroundings open up, and to 175 before Lance. The last big step comes in Kanto, where 40 more species appear, and Mt. Silver adds the last four, including the Larvitar line, only in time for Red.

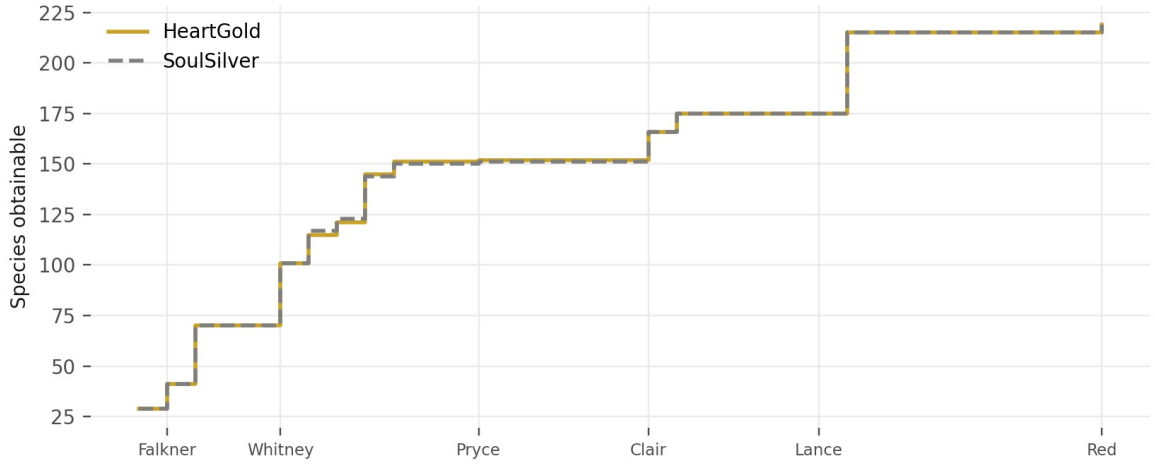


Figure 1. Number of species a player can have in the team before each boss battle, in a single cartridge without trading, legendary or event Pokémon. Figure produced using Matplotlib (Hunter, 2007).

Moves open up more slowly than Pokémon. Of the 100 TMs and HMs, 84 can be obtained outside the Game Corner, but the best special attacks come late: before the Johto league a player can buy Fire Blast, Thunder and Blizzard in Goldenrod, and Earthquake lies in Victory Road, but Ice Beam only appears in Kanto's Seafoam Islands, and Thunderbolt and Flamethrower only in Cerulean Cave and on Route 28, which require all sixteen badges. Held items follow the same pattern. Before Whitney a player can collect Mystic Water in Cherrygrove, Charcoal in Azalea, a Hard Stone and a TwistedSpoon from two of the seven siblings who each give an item on one day of the week, and renewable Sitrus Berries; Leftovers only come from the Snorlax in Vermilion City, once the story reaches Kanto.

4.2 The best team for the story

Table 1 reports the best team found for each starter in each version, among the 15 teams that reached the final round and were played through the story battle by battle. The winner is the Totodile team in HeartGold: Feraligatr, Arcanine, Graveler, Heracross, Lapras and Umbreon. In 40-battle runs it only has to be levelled above a boss's level once, for the Elite Four and Lance, and on average it fights two levels below its opponents. It is the same team that came first in the second stage, when moves were chosen afresh for every battle, so the result does not depend on moves the team could not yet have had. In SoulSilver, where Growlithe does not exist, the best Totodile team replaces Arcanine with Crobat. The best Cyndaquil and Chikorita teams need five to eight levels of training above their opponents in total, depending on the version.

Table 1. The best team found for each starter and version.

Version	Starter	Other five members	Training above boss's level	Battles that need it	Average level vs boss	Level for Blue
HeartGold	Chikorita	Graveler, Heracross, Starmie, Steelix, Umbreon	5	Clair, Elite Four	-0.6	58
HeartGold	Cyndaquil	Graveler, Gyarados, Heracross, Lapras, Umbreon	5	Rival, Elite Four	-0.8	55
HeartGold	Totodile	Arcanine, Graveler, Heracross, Lapras, Umbreon	1	Elite Four	-2.2	58
SoulSilver	Chikorita	Graveler, Gyarados, Heracross, Lapras, Umbreon	8	Bugsy, Elite Four	-0.7	56
SoulSilver	Cyndaquil	Graveler, Heracross, Lapras, Steelix, Umbreon	6	Rival, Morty, Elite Four	-0.6	58
SoulSilver	Totodile	Crobat, Graveler, Heracross, Lapras, Umbreon	7	Bugsy, Elite Four	-0.2	60

Note: Each team plays the 34 story battles in order, with the moves, TMs and held items it can really have at each point (Section 3.4), and its level never goes down. For each battle the team is levelled up only if it wins less than 90% of 40 simulated battles at its current level. 'Training above boss's level' adds up, over the battles for which the team had to be levelled up, the levels by which it then exceeds the boss's highest level; 'Average level vs boss' is the mean of team level minus boss level over the 34 battles. Arcanine is not obtainable in SoulSilver. Source: Author's own calculations.

The most striking result is not the winner but what all the good teams have in common. Graveler and Umbreon appear in every one of the 15 finalists, Heracross in 13 and Lapras in 12. Each fills a gap that the starters leave open. Graveler, a Geodude caught on Route 46 in the first minutes of the game, is a Rock and Ground type that shrugs off the Flying, Bug, Fire, Electric and Poison attacks of Falkner, Bugsy, Jasmine's Magnemite, the Rocket executives and Koga, and hits most of them super-effectively. Heracross, found in Headbutt trees once the Headbutt tutor in Ilex Forest is met, brings Fighting moves, the best answer to Whitney's Miltank, Pryce's Ice types, Karen's Dark types and every Normal and Rock type in Kanto. Umbreon, evolved from the Eevee Bill gives away in Goldenrod, is one of the bulkiest Pokémon available and its Dark moves handle Morty's Ghosts and the Psychic types of Will and Sabrina. Lapras, the

Friday encounter in Union Cave, carries Ice attacks into the Dragon battles against Clair and Lance.

Table 2 shows the team as it faces Lance, at the end of the Elite Four: each Pokémon's ability, held item and four moves. Appendix C gives the same information for every one of the 34 story battles.

Table 2. The best team against Lance (level 54): abilities, held items and moves.

Pokémon	Ability	Held item	Moves
Feraligatr	Torrent	Sitrus Berry	Ice Fang, Crunch, Earthquake, Waterfall
Arcanine	Intimidate	Sitrus Berry	Fire Blast, Thunder Fang, Fire Fang, Dragon Pulse
Graveler	Rock Head	Hard Stone	Fire Blast, Stone Edge, Rollout, Earthquake
Heracross	Swarm	Black Belt	Close Combat, Strength, Aerial Ace, Shadow Claw
Lapras	Water Absorb	Sitrus Berry	Ice Beam, Surf, Waterfall, Thunder
Umbreon	Synchronize	BlackGlasses	Iron Tail, Dark Pulse, Feint Attack, Hyper Beam

Note: Neutral nature, average individual values (15), no effort values. Win rate at this level: 90.5% of 400 battles. The moves and items for all 34 battles are in Appendix C. Source: Author's own calculations.

4.3 Assembling the team, battle by battle

Table 3 gives the order in which the team comes together, and Table 4 the TMs and HMs it uses. The first two members are available before the first battle: the starter and a Geodude from Route 46, a Pokémon with the maximum catch rate. Growlithe, on Route 36, joins before Bugsy and becomes Arcanine before Whitney, which requires a Fire Stone from the Pokéathlon Dome on a Tuesday. Heracross joins for Whitney, once the Headbutt tutor in Ilex Forest has been met; it is found at level 2 or 3 and must be raised to the team's level, 18. Bill's Eevee, received in Goldenrod, is evolved into Umbreon with high friendship at night before Morty. Lapras, the last member, is caught on a Friday in Union Cave once Surf is available, and joins before Chuck. With a catch rate of 45, it is the only member for which the ball matters: an Ultra Ball thrown at full HP succeeds only 12% of the time, but at low HP on a sleeping target the probability rises to 78%, so it is worth arriving with a Pokémon that can put it to sleep.

Table 3. How to assemble the best team, in order (HeartGold, Totodile).

No.	Get	Where	Level found	First battle	Team level then	Evolutions	Catch rate	Ultra Ball, low HP asleep
1	Totodile	starter	5	Rival	5	Croconaw before Whitney (level 18); Feraligatr before Clair (level 38)	–	–
2	Geodude	Route 46 (land)	2	Rival	5	Graveler before Chuck (level 28)	255	100%
3	Growlithe	Route 36 (land)	13	Bugsy	15	Arcanine before Whitney (level 18)	190	100%

No.	Get	Where	Level found	First battle	Team level then	Evolutions	Catch rate	Ultra Ball, low HP asleep
4	Heracross	Headbutt trees, Route 46	2	Whitney	18	does not evolve	45	78%
5	Eevee	gift from Bill (Goldenrod)	5	Morty	20	Umbreon before Morty (level 20)	–	–
6	Lapras	static encounter (Union Cave, Fridays)	20	Chuck	28	does not evolve	45	78%

Note: 'First battle' is the first boss battle in which the Pokémon takes part; it has to be trained from the level at which it is found to the team level shown. Evolutions happen as soon as they can: by level, with a Fire Stone (Pokéathlon Dome, Tuesdays) for Arcanine and by friendship at night for Umbreon. Catch probability per Ultra Ball thrown at 1/30 of maximum HP on a sleeping target, from the game's integer catch formula. Source: Author's own calculations.

Early on, the team's strength comes from the Goldenrod department store. Before Whitney the plan buys Blizzard for Croconaw, Fire Blast and Solar Beam for Arcanine, a second Fire Blast for Geodude and Focus Blast for Heracross, 25,000 in total, and over the whole story the TMs it buys cost 91,500, since a TM is used up every time it is taught. The single-use TMs are few but important: Rock Tomb for Totodile against Proton's Zubat and Koffing, Payback, Iron Tail and later Dark Pulse and Psychic for Umbreon, Earthquake and Ice Beam for Feraligatr, and Dragon Pulse for Arcanine against Lance's Dragonite. Surf and Strength must be taught before Chuck, and because an HM move cannot be forgotten before the Move Deleter in Blackthorn City, Rock Smash and Strength stay in several movesets until Clair.

Table 4. Which TMs and HMs to teach, to whom and when (best team).

Before	TM/ HM	Move	Teach to	Where to get it
Proton (Slowpoke Well)	TM39	Rock Tomb	Totodile	item ball: Union Cave B1F
Whitney	TM14	Blizzard	Feraligatr	Goldenrod Dept. Store 5F, bought for 5,500
Whitney	TM22	Solar Beam	Arcanine	Goldenrod Dept. Store 5F, bought for 3,000
Whitney	TM28	Dig	Geodude	item ball: National Park
Whitney	TM38	Fire Blast	Arcanine	Goldenrod Dept. Store 5F, bought for 5,500
Whitney	TM38	Fire Blast	Graveler	Goldenrod Dept. Store 5F, bought for 5,500
Whitney	TM52	Focus Blast	Heracross	Goldenrod Dept. Store 5F, bought for 5,500
Rival (Burned Tower)	HM06	Rock Smash	Feraligatr	Route 36 (after Sudowoodo)
Rival (Burned Tower)	HM06	Rock Smash	Heracross	Route 36 (after Sudowoodo)
Morty	HM06	Rock Smash	Graveler	Route 36 (after Sudowoodo)
Morty	TM15	Hyper Beam	Umbreon	Goldenrod Dept. Store 5F, bought for 7,500
Morty	TM66	Payback	Umbreon	item ball: Route 35
Chuck	HM03	Surf	Feraligatr	Kimono Girls, Ecruteak (after Morty)
Chuck	HM03	Surf	Lapras	Kimono Girls, Ecruteak (after Morty)
Chuck	HM04	Strength	Feraligatr	Route 42

Before	TM/ HM	Move	Teach to	Where to get it
Chuck	HM04	Strength	Arcanine	Route 42
Chuck	HM04	Strength	Graveler	Route 42
Chuck	HM04	Strength	Heracross	Route 42
Chuck	TM14	Blizzard	Lapras	Goldenrod Dept. Store 5F, bought for 5,500
Chuck	TM25	Thunder	Lapras	Goldenrod Dept. Store 5F, bought for 5,500
Ariana (Rocket Hideout)	HM06	Rock Smash	Lapras	Route 36 (after Sudowoodo)
Ariana (Rocket Hideout)	TM23	Iron Tail	Umbreon	Olivine Gym
Pryce	TM65	Shadow Claw	Heracross	item ball: Route 42
Rival (Goldenrod Underground)	HM06	Rock Smash	Arcanine	Route 36 (after Sudowoodo)
Ariana (Radio Tower)	TM14	Blizzard	Feraligatr	Goldenrod Dept. Store 5F, bought for 5,500
Archer (Radio Tower)	TM22	Solar Beam	Arcanine	Goldenrod Dept. Store 5F, bought for 3,000
Clair	HM07	Waterfall	Feraligatr	item ball: Ice Path 1F
Clair	HM07	Waterfall	Lapras	item ball: Ice Path 1F
Rival (Victory Road)	TM26	Earthquake	Feraligatr	item ball: Victory Road 2F
Rival (Victory Road)	TM79	Dark Pulse	Umbreon	item ball: Victory Road 3F
Elite Four (Will)	TM15	Hyper Beam	Umbreon	Goldenrod Dept. Store 5F, bought for 7,500
Elite Four (Will)	TM25	Thunder	Lapras	Goldenrod Dept. Store 5F, bought for 5,500
Elite Four (Will)	TM59	Dragon Pulse	Arcanine	Blackthorn Gym
Brock	TM13	Ice Beam	Feraligatr	item ball: Seafoam Islands B4F
Brock	TM22	Solar Beam	Arcanine	Goldenrod Dept. Store 5F, bought for 3,000
Brock	TM28	Dig	Umbreon	Celadon Dept. Store 3F, bought for 2,000
Brock	TM29	Psychic	Umbreon	Saffron Mr Psychic House
Brock	TM52	Focus Blast	Feraligatr	Goldenrod Dept. Store 5F, bought for 5,500
Misty	TM80	Rock Slide	Heracross	Pewter Gym
Erika	TM28	Dig	Arcanine	Celadon Dept. Store 3F, bought for 2,000
Sabrina	TM25	Thunder	Lapras	Goldenrod Dept. Store 5F, bought for 5,500
Rival (Mt. Moon)	TM50	Overheat	Arcanine	Seafoam Islands Cinnabar Gym
Blue	TM22	Solar Beam	Arcanine	Goldenrod Dept. Store 5F, bought for 3,000
Blue	TM25	Thunder	Lapras	Goldenrod Dept. Store 5F, bought for 5,500

Note: TMs found once (gifts, gym rewards, items on the ground) are used up when taught; those sold in the Goldenrod and Celadon department stores can be bought again. HM moves cannot be forgotten before the Move Deleter in Blackthorn City. Source: Author's own calculations.

Held items change more often than moves. Table 5 lists where each item is found and which Pokémon holds it in which battles; over the 34 battles the plan changes a member's item 65 times. A few patterns are stable. Graveler holds a Hard Stone or, from Blackthorn onwards, a Soft Sand; Arcanine holds Charcoal against many trainers; Umbreon holds BlackGlasses from Clair onwards; Lapras holds Mystic Water, Heracross a Black Belt or, in Kanto, Leftovers.

Every other member holds a Sitrus Berry, which is the most useful item in the game for this purpose because it can be picked again and held by several Pokémon at once. The items make a real difference: played without them, the same team needs level 57 for the Elite Four instead of 54.

Table 5. Held items: where to find them and who holds them in which battles (best team).

Item	Where	Held by	Battles
Mystic Water	Cherrygrove City (gift)	Feraligatr	Falkner, Archer (Radio Tower)
		Lapras	Chuck to Jasmine, Ariana (Rocket Hideout), Petrel (Radio Tower), Koga to Bruno, Lt. Surge, Janine to Sabrina
Sitrus Berry	berry trees (renewable)	Feraligatr	Proton (Slowpoke Well) to Chuck, Petrel (Rocket Hideout) to Pryce, Proton (Radio Tower) to Ariana (Radio Tower), Rival (Victory Road) to Erika, Sabrina to Blue
		Arcanine	Bugsy to Rival (Azalea), Morty, Clair to Koga, Karen to Lance, Misty, Rival (Mt. Moon) to Blue
		Heracross	Whitney to Morty, Jasmine to Ariana (Rocket Hideout), Petrel (Radio Tower) to Karen
		Graveler	Rival (Burned Tower) to Rival (Goldenrod Underground), Rival (Victory Road) to Karen, Lt. Surge to Erika, Blaine to Blue
		Umbreon	Morty to Archer (Radio Tower), Rival (Victory Road) to Will, Bruno to Karen, Sabrina, Rival (Mt. Moon)
		Lapras	Petrel (Rocket Hideout), Rival (Goldenrod Underground) to Will, Karen to Brock, Blaine to Blue
Hard Stone	Route 36, from Arthur (Thursdays)	Graveler	Falkner to Whitney, Proton (Radio Tower) to Archer (Radio Tower), Lance, Janine
TwistedSpoon	Route 29, from Tuscany (Tuesdays)	Umbreon	Janine
Charcoal	Azalea Town, charcoal kiln (gift)	Arcanine	Whitney to Rival (Burned Tower), Chuck to Ariana (Radio Tower), Bruno, Lt. Surge to Erika
Magnet	Route 37, from Sunny (Sundays)	Lapras	Pryce, Misty
Black Belt	Lake of Rage, from Wesley (Wednesdays)	Heracross	Pryce, Lance, Erika
		Arcanine	Archer (Radio Tower)
Sharp Beak	Route 40, from Monica (Mondays)	Heracross	Chuck
Wave Incense	item ball: Route 47	Feraligatr	Jasmine, Petrel (Radio Tower) to Rival (Goldenrod Underground), Janine
BlackGlasses	Dark Cave	Umbreon	Clair, Koga, Lance to Erika, Blaine, Blue
		Arcanine	Sabrina
Soft Sand	Blackthorn City, from Santos (Saturdays)	Graveler	Clair, Brock to Misty, Sabrina
		Arcanine	Blaine
Never-Melt Ice	Ice Path	Feraligatr	Clair

Item	Where	Held by	Battles
Dragon Fang	item ball: Dragon's Den	Arcanine	Brock
Leftovers	held by the Vermilion City Snorlax	Heracross	Brock to Lt. Surge, Sabrina to Blue
		Lapras	Erika
		Arcanine	Janine
Rock Incense	item ball: Diglett Cave	Heracross	Janine

Note: Items are swapped freely between battles; each copy is held by one Pokémon at a time. Sitrus Berries are renewable and can be held by several Pokémon at once. Items are assigned before each battle by their expected benefit against that trainer's team. Source: Author's own calculations.

4.4 Where the walls are

Figure 2 shows, for the best team of each starter, the team's level in the plan compared with each boss's level. For most of the story the curve sits below zero. Having trained to level 28 for Chuck, the Totodile team then wins every battle up to Archer in the Radio Tower, Pryce included, six to ten levels below its opponents, because the Rocket executives and gym leaders of that stretch field few, mono-type teams that the core covers easily. The curve rises above zero only once, at the Elite Four, and in Kanto it is only above a gym leader's level because of the level reached for Lance. Table 6 gives the level and the win rate at four key battles, with 400 battles per level.

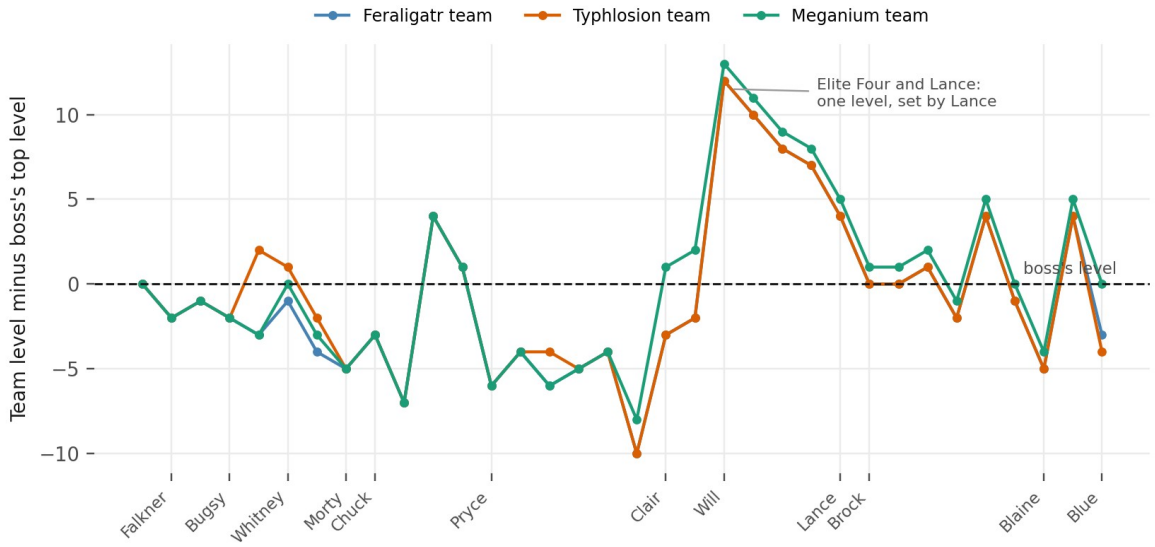


Figure 2. Team level in the plan minus the boss's highest level, for the best HeartGold team of each starter in all 34 story battles (400 battles per level). The team's level never goes down, and the Elite Four and Lance are fought at one level. Figure produced using Matplotlib (Hunter, 2007).

Table 6. Team level in the plan at four key battles, with the win rate (400 battles per level).

Team	Whitney (19)	Clair (41)	Lance (50)	Blue (60)
Feraligatr team	18 (-1), 92%	38 (-3), 90%	54 (+4), 90%	57 (-3), 91%

Team	Whitney (19)	Clair (41)	Lance (50)	Blue (60)
Typhlosion team	20 (+1), 99%	38 (-3), 93%	54 (+4), 94%	56 (-4), 94%
Meganium team	19 (+0), 98%	42 (+1), 91%	55 (+5), 90%	60 (+0), 97%
Feraligatr team, no held items	18 (-1), 91%	40 (-1), 99%	57 (+7), 100%	60 (+0), 94%

Note: HeartGold, best team of each starter in Table 1, re-played with 400 battles per level. Boss's highest level in brackets in the header; in the cells, team level, difference from the boss's level, and win rate. The team level is the lowest at which the plan wins at least 90% of the time, but never below its level in the previous battle, so a team can be above a boss's level because of an earlier, harder battle. Source: Author's own calculations.

Whitney is not one of the walls for a good team: all three teams beat her within one level of her Miltank. Clair is close to the line, and Blue is below it: the Totodile team beats him three levels under his own. The one stage that forces every team to train is the Elite Four. Lance's team of Gyarados, Aerodactyl, Charizard and three Dragonite, the last holding a Sitrus Berry, is backed by four Full Restores, and because the five battles are fought in a row with no chance to leave and train, the team must enter at the level Lance requires: 54 or 55 for all three teams, four or five levels above his strongest Pokémon. Will, Koga, Bruno and Karen are then won almost every time.

Miltank's reputation comes from somewhere else. Figure 3 repeats the exercise with the starter alone, without the rest of the team and with moves chosen afresh for each battle, as in the second stage. On its own, every starter needs 8 to 11 extra levels to beat Whitney, the first battle in which all three need at least eight, and Chikorita needs level 38 to beat Bugsy's level-17 Scyther. From Clair onwards a starter alone is hopeless: Chikorita cannot win seven of the story battles even 40 levels above its opponents, and no starter can beat Lance alone. The difficulty that players remember is, in large part, the difficulty of relying on one Pokémon.

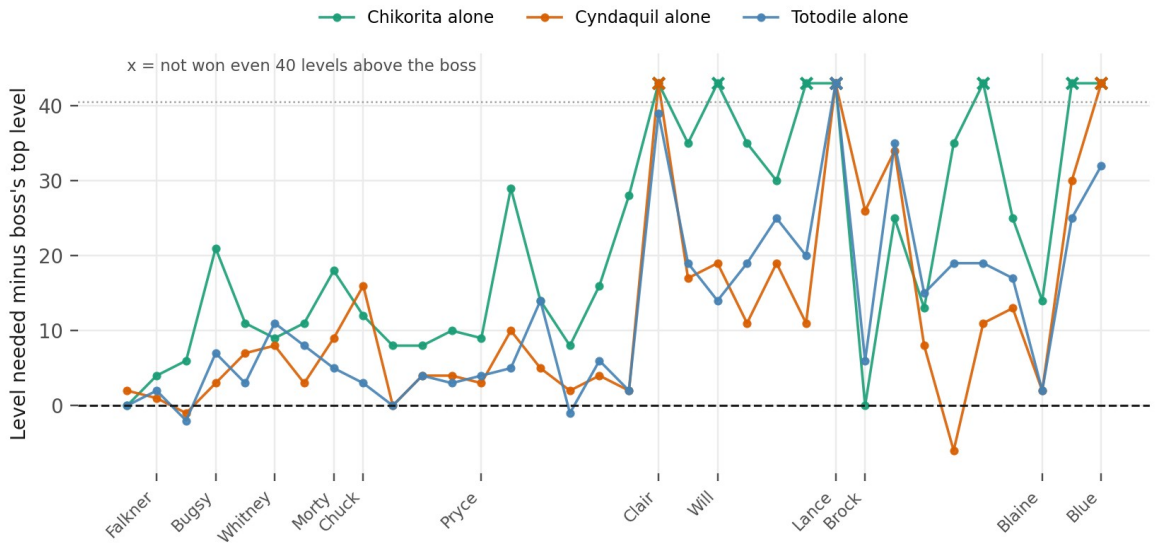


Figure 3. Level needed for the starter alone to win at least 90% of battles, minus the boss's highest level (HeartGold, 40 battles per level). Crosses mark battles not won even 40 levels above the boss. Figure produced using Matplotlib (Hunter, 2007).

4.5 Beating Red

Red is in a different league from anything in the story. His Pokémon are level 80 to 88, three of them are the fully evolved Kanto starters, and he carries four Full Restores. The story team, trained up, is not the right tool: the screen for Red selects a different team entirely. Table 7 lists the four best endgame teams, now with held items, including those bought with Battle Points at the Battle Frontier. All share five members, Blissey, Tyranitar, Steelix, Tentacruel and Feraligatr, and differ only in the sixth; they are within three levels of each other, which is close to the precision of 40-battle runs. The best, with Ampharos as the sixth member, needs level 84: a re-simulation with 400 battles gives 81% wins at level 82, 87% at 83 and 91.5% at 84. Figure 4 shows how steeply the win rate rises around Red's own level.

Table 7. The best endgame teams: level needed to win 90% of battles against Red and the Elite Four rematches.

Team	Red (88)	Will (62)	Koga (64)	Bruno (64)	Karen (64)	Lance (75)
Ampharos, Blissey, Feraligatr, Steelix, Tentacruel, Tyranitar	83	55	55	72	62	79
Blissey, Feraligatr, Slaking, Steelix, Tentacruel, Tyranitar	84	55	55	78	60	77
Blissey, Electrode, Feraligatr, Steelix, Tentacruel, Tyranitar	84	55	56	75	62	79
Blissey, Feraligatr, Nidoking, Steelix, Tentacruel, Tyranitar	86	58	55	71	59	82

Note: Every obtainable Pokémon allowed (HeartGold), with held items including those bought at the Battle Frontier; TMs already used up by the story team are not available. Highest level of each trainer's team in brackets. Levels from 40 battles per level; for the first team against Red, 400 battles give 81.0% at level 82; 87.0% at level 83; 91.5% at level 84. Source: Author's own calculations.

Each member has a clear role. Blissey, with the highest HP in the game and a huge Special Defense, absorbs the special attacks of Lapras, Venusaur and Blastoise and hits back with Thunder, Fire Blast and Focus Blast. Steelix is immune to Pikachu's Electric attacks and to Venusaur's Sludge Bomb, and resists Red's Normal, Ice, Psychic, Ghost and Dark attacks. Tyranitar's Sand Stream summons a sandstorm that chips away at every one of Red's Pokémon, none of which is a Rock, Ground or Steel type, and Tentacruel resists the Water, Ice and Fire attacks that make up much of Red's offence. Ampharos adds Thunderbolt against Lapras, Blastoise and Charizard. The weak spot of these teams is Fighting moves, which is why Bruno's rematch team, with Hitmonlee, Machamp and Lucario, requires level 71 to 78 even though his strongest Pokémon is only level 64. Table 8 lists the abilities, items and moves of the best team against Red; the TMs the story team used up, such as Earthquake and Psychic, are not available to it.

Table 8. The best endgame team against Red (level 84): abilities, held items and moves.

Pokémon	Ability	Held item	Moves
Steelix	Rock Head	Muscle Band	Stone Edge, Double-Edge, Fire Fang, Thunder Fang
Ampharos	Static	Leftovers	Thunderbolt, Fire Punch, Focus Blast, Discharge
Tyranitar	Sand Stream	Expert Belt	Stone Edge, Crunch, Focus Punch, Fire Blast
Feraligatr	Torrent	Sitrus Berry	Superpower, Aqua Tail, Ice Beam, Waterfall
Tentacruel	Clear Body	Black Sludge	Surf, Sludge Bomb, Blizzard, Poison Jab
Blissey	Natural Cure	Wise Glasses	Hyper Beam, Thunder, Fire Blast, Focus Blast

Note: Neutral nature, average individual values (15), no effort values. Choice Band, Choice Scarf, Scope Lens, Focus Band, Muscle Band and BrightPowder cost 48 Battle Points each at the Battle Frontier. Source: Author's own calculations.

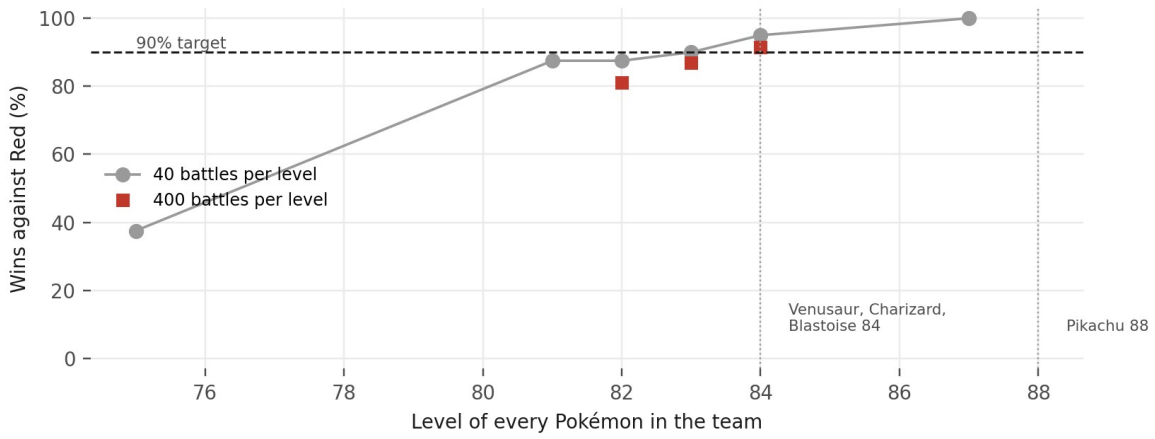


Figure 4. Win rate against Red by team level for the best endgame team (HeartGold, with held items). Figure produced using Matplotlib (Hunter, 2007).

The endgame team, listed in Table 9, is harder to assemble than the story team. Chansey, which evolves into Blissey with high friendship, appears only in Kanto, for example on Route 13, and has a catch rate of 30; Steelix, caught directly in Cliff Cave, has a catch rate of 25; and Larvitar lives only in Mt. Silver, the last area of the game, from where it must be raised to level 55 to become Tyranitar.

Table 9. Where to get the endgame team.

Pokémon	Line	Where	Available before	Level	Catch rate	Ultra Ball, low HP asleep
Ampharos	Mareep → Flaaffy → Ampharos	Route 32 (land)	Proton	6	235	100%
Blissey	Chansey → Blissey	Route 13 (land)	Brock	25	30	50%
Feraligatr	Totodile → Croconaw → Feraligatr	starter	Rival	5	–	–
Steelix	Steelix	Cliff Cave (land)	Jasmine	23	25	41%
Tentacruel	Tentacool → Tentacruel	New Bark Town (Old Rod)	Proton	10	190	100%
Tyranitar	Larvitar → Pupitar → Tyranitar	Mt. Silver, inner cave (land)	Red	15	45	78%

Note: As in Table 3. Source: Author's own calculations.

4.6 The fastest way to train

Because experience depends only on the defeated Pokémon, the ranking of training spots is simple: the more high-level, high-yield Pokémon a spot offers per minute, the better. Table 10 lists the two best repeatable sources at each stage. Before Whitney, Union Cave's lowest floor, with wild Pokémon of levels 20 to 23, gives about 630 experience points per minute. Before Clair, Mt. Mortar gives about 1,090 and, before the Elite Four, Victory Road about 1,420. After beating Lance for the first time, rebattling the Elite Four with their original teams gives 2,400 per minute, and once all sixteen badges are won their rematch teams, between levels 58 and 75, give almost 4,000 per minute, about 75% more than the best spot on Mt. Silver.

Table 10. The best repeatable sources of experience at each stage of the game.

Stage	Source	Opponent levels	Experience per battle or run	Experience per minute
Before Whitney	Union Cave B2F (land)	20-23	315	630
	Ruins of Alph (land)	18-24	225	450
Before Clair	Mount Mortar 2F (land)	28-32	544	1,088
	Route 47 (land)	31-40	511	1,022
Before Lance	Victory Road 1F (land)	32-36	710	1,420
	Route 26 (land)	28-32	549	1,099
After Lance (Kanto)	Elite Four + Lance, first team (Johto league)	40-50	42,513	2,406
	Seafoam Islands B4F (super rod)	40-40	978	1,957
All 16 badges	Elite Four + Lance rematch (full run)	58-75	75,205	3,958
	Mount Silver Cave 1F (super rod)	40-40	1,132	2,263

Note: Experience for one Pokémon taking part alone, without Lucky Egg. Wild spots use daytime encounter tables and slot probabilities; a wild encounter is assumed to take 30 seconds, a trainer battle 60 seconds plus 20 seconds per Pokémon, and an Elite Four run 4 extra minutes. The Fighting Dojo gym leader rematches give even more per minute but only once a day on set days, so they are not listed as repeatable. Source: Author's own calculations.

Table 11 turns these rates into a training budget for the plan, under the pessimistic assumption that every point of experience had to come from grinding rather than from the story's trainers. Most of the early steps take minutes. The two real budgets are before Clair, where the team goes from level 28 to 38 in about three hours at Mt. Mortar, and above all before the Elite Four, where raising six Pokémon from level 38 to 54 takes about 690,000 experience points, or eight hours at Victory Road, much of which the story's trainers provide along the way. The whole story, from the first battle to Blue, adds up to at most about 15 hours of training. Raising the endgame team from level 60 to 84 takes about 2.4 million points, or ten hours of Elite Four rematches; a Lucky Egg multiplies the holder's experience by 1.5, which would cut the time for that Pokémon by a third.

Table 11. When to train and how long it takes, if all the experience came from the best source.

Before	Team level	Members	Best source available	Experience needed (team)	Hours
Falkner	5 → 11	2	Route 36 (land)	1,214	0.1
Bugsy	11 → 15	3	Union Cave B2F (land)	4,058	0.1
Whitney	15 → 18	4	Union Cave B2F (land)	13,878	0.4
Morty	18 → 20	5	Union Cave B2F (land)	16,619	0.4
Chuck	20 → 28	6	Mount Mortar 2F (land)	89,836	1.4
Clair	28 → 38	6	Mount Mortar 2F (land)	217,578	3.3
Elite Four and Lance	38 → 54	6	Victory Road 1F (land)	692,572	8.1
Blue	54 → 57	6	Elite Four + Lance, first team (Johto league)	188,872	1.3
Red (endgame team)	60 → 84	6	Elite Four + Lance rematch (full run)	2,425,035	10.2

Note: Upper bounds. One row for each battle before which the plan raises the team's level (verified with 400 battles). Experience counts every member from its current level, or from the level at which it is found; experience earned from trainers along the way reduces the time. The endgame row raises the six endgame Pokémon from level 60. Source: Author's own calculations.

5. What It Means

5.1 A practical guide

The results translate into a simple plan, which Appendix C spells out battle by battle. Choose Totodile if there is a choice of starter. Before the first rival battle, catch a Geodude on Route 46: it is the single most useful Pokémon of the early game, and it holds a Hard Stone from Arthur on Route 36 on Thursdays. Pick up the Rock Tomb TM in Union Cave for Totodile before Proton's battle in the Slowpoke Well, and catch a Growlithe on Route 36 before Bugsy. After Bugsy, learn Headbutt in Ilex Forest, shake the trees for a Heracross, collect the Charcoal in Azalea, buy a Fire Stone at the Pokéathlon on a Tuesday, and spend the savings in the Goldenrod department store: Blizzard, Fire Blast, Solar Beam and Focus Blast. At level 18 the team beats Whitney nine times out of ten.

Collect Bill's Eevee in Goldenrod and evolve it into Umbreon at night before Morty, with Payback from Route 35. Once Surf is available, catch Lapras in Union Cave on a Friday and teach it Surf, Blizzard and Thunder. At level 28 the team then beats Chuck and everything up to the Radio Tower. Train to 38 at Mt. Mortar for Clair, pick up Waterfall in the Ice Path and use the Move Relearner in Blackthorn to recover the level-up moves dropped along the way. Train to 54 in Victory Road, where Earthquake and Dark Pulse lie on the ground, before entering the Elite Four. Kanto then needs no further training until Blue, at 57.

For Red, it pays to build a second team: Chansey from Kanto's Route 13, Steelix from Cliff Cave, Larvitar from Mt. Silver, Tentacool from any fishing spot, Mareep from Route 32 and the starter, trained to level 84 on the Elite Four rematches.

5.2 What the model leaves out

The simulations make several simplifications, and each deserves a caveat. First, both sides follow simple rules. The simulated player never uses items in battle, never uses status moves and rarely switches, which makes the required levels conservative, since a real player with a few Potions or a Sleep Powder does better. The simulated trainers use their best attack four times out of five and a random move otherwise, which is close to, but not exactly, the game's own battle AI. Second, the player's Pokémon have average individual values and no effort values. Moves and held items are chosen by a simple rule, the expected damage against the next trainers, rather than by testing every combination, so a better plan may exist. Third, each Elite Four member is fought with a fully healed team, whereas in the game the player carries damage between them and can only heal with items. Fourth, money is not modelled: the plan's TMs cost 91,500 and some of them are bought early, which may require rematches with trainers, and field moves such as Cut, Flash and Whirlpool are assumed to be carried by a Pokémon kept in the PC. Fifth, some ways to obtain Pokémon were left out: the Safari Zone's object-dependent encounters, swarms, the radio sounds, the Pokéwalker and breeding. Finally,

the levels come from simulated battles and are uncertain by about one level: the ranking of the 15 teams uses 40 battles per level, and the levels reported for the best teams use 400.

6. Conclusion

This paper set out to find, with numbers rather than folklore, the best team to beat Pokémon HeartGold and SoulSilver, where to catch it, what it should know and hold, and how to train it. Using the game's own data and several hundred thousand simulated battles, the answer is a team built around four Pokémon that a player can collect before the fifth gym: Graveler, Heracross, Umbreon and Lapras. With Totodile as the starter and Arcanine in HeartGold, or Crobat in SoulSilver, this team only needs to out-level its opponents for the Elite Four, and it does so with the moves and items a player really has at each point of the story.

Two popular beliefs come out differently. Whitney's Miltank is a real obstacle, but for a starter fighting alone; with a balanced team she is an ordinary gym leader. The game's hardest story battle is Lance, not Whitney, and Red is harder still, requiring a purpose-built team of Blissey, Tyranitar, Steelix and Tentacruel at level 84. The best way to get there is also the least romantic: not the snowy caves of Mt. Silver, but repeated runs through the Elite Four.

The same approach could answer many other questions about the game, from the best team for a player who refuses to train above each boss's level to the fastest route through the story. The data, the simulator and every script used here are provided with the paper for anyone who wants to try.

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Appendixes

Appendix A. The best team (HeartGold, Totodile) in all 34 story battles.

Battle	Boss's top level	Team level	Difference	Wins (400 battles)	95% interval
Rival (Cherrygrove)	5	5	+0	100.0%	99.0–100.0%
Falkner	13	11	-2	100.0%	99.0–100.0%
Proton (Slowpoke Well)	12	11	-1	100.0%	99.0–100.0%
Bugsy	17	15	-2	91.8%	88.6–94.1%
Rival (Azalea)	18	15	-3	93.0%	90.1–95.1%
Whitney	19	18	-1	92.5%	89.5–94.7%
Rival (Burned Tower)	22	18	-4	100.0%	99.0–100.0%
Morty	25	20	-5	94.5%	91.8–96.3%
Chuck	31	28	-3	97.2%	95.1–98.5%
Jasmine	35	28	-7	100.0%	99.0–100.0%
Petrel (Rocket Hideout)	24	28	+4	100.0%	99.0–100.0%
Ariana (Rocket Hideout)	27	28	+1	100.0%	99.0–100.0%
Pryce	34	28	-6	100.0%	99.0–100.0%
Petrel (Radio Tower)	32	28	-4	100.0%	99.0–100.0%
Rival (Goldenrod Underground)	34	28	-6	99.5%	98.2–99.9%
Proton (Radio Tower)	33	28	-5	100.0%	99.0–100.0%
Ariana (Radio Tower)	32	28	-4	100.0%	99.0–100.0%
Archer (Radio Tower)	38	28	-10	99.2%	97.8–99.7%
Clair	41	38	-3	90.0%	86.7–92.6%
Rival (Victory Road)	40	38	-2	100.0%	99.0–100.0%
Will	42	54	+12	100.0%	99.0–100.0%
Koga	44	54	+10	100.0%	99.0–100.0%
Bruno	46	54	+8	100.0%	99.0–100.0%
Karen	47	54	+7	100.0%	99.0–100.0%
Lance	50	54	+4	90.5%	87.2–93.0%
Brock	54	54	+0	100.0%	99.0–100.0%
Misty	54	54	+0	100.0%	99.0–100.0%
Lt. Surge	53	54	+1	100.0%	99.0–100.0%
Erika	56	54	-2	100.0%	99.0–100.0%
Janine	50	54	+4	100.0%	99.0–100.0%
Sabrina	55	54	-1	100.0%	99.0–100.0%
Blaine	59	54	-5	100.0%	99.0–100.0%
Rival (Mt. Moon)	50	54	+4	99.8%	98.6–100.0%
Blue	60	57	-3	91.0%	87.8–93.4%

Note: Team level = lowest level at which the plan wins at least 90% of 400 battles, never below the level of the previous battle. Wilson 95% confidence intervals. Source: Author's own calculations.

Appendix B. Every team in the final round, played battle by battle.

Version	Team	Training above boss's level	Average level vs boss	Level for Blue	Average win rate
HeartGold	Feraligatr, Arcanine, Graveler, Heracross, Lapras, Umbreon	1	-2.2	58	97%
HeartGold	Feraligatr, Arcanine, Exeggutor, Graveler, Lapras, Umbreon	1	-1.0	59	98%

Version	Team	Training above boss's level	Average level vs boss	Level for Blue	Average win rate
HeartGold	Feraligatr, Arcanine, Espeon, Graveler, Lapras, Umbreon	2	-1.8	56	98%
HeartGold	Typhlosion, Graveler, Gyarados, Heracross, Lapras, Umbreon	5	-0.8	55	98%
HeartGold	Typhlosion, Ampharos, Graveler, Heracross, Lapras, Umbreon	5	-0.8	59	98%
HeartGold	Meganium, Graveler, Heracross, Starmie, Steelix, Umbreon	5	-0.6	58	98%
HeartGold	Typhlosion, Graveler, Heracross, Lapras, Nidoqueen, Umbreon	5	-0.3	60	98%
SoulSilver	Typhlosion, Graveler, Heracross, Lapras, Steelix, Umbreon	6	-0.6	58	98%
HeartGold	Meganium, Graveler, Heracross, Lapras, Steelix, Umbreon	7	-0.7	63	97%
SoulSilver	Feraligatr, Crobat, Graveler, Heracross, Lapras, Umbreon	7	-0.2	60	99%
SoulSilver	Meganium, Graveler, Gyarados, Heracross, Lapras, Umbreon	8	-0.7	56	99%
SoulSilver	Meganium, Graveler, Heracross, Steelix, Umbreon, Vaporeon	8	+0.4	61	97%
HeartGold	Meganium, Graveler, Gyarados, Heracross, Steelix, Umbreon	10	+0.1	64	98%
SoulSilver	Typhlosion, Graveler, Heracross, Jolteon, Lapras, Umbreon	10	+0.9	58	99%
SoulSilver	Feraligatr, Beedrill, Graveler, Heracross, Lapras, Umbreon	10	+0.9	62	98%

Note: 34 story battles each, 40 battles per level, ranked by the first column and then the second. See Table 1. Source: Author's own calculations.

Appendix C. The complete plan: moves and held item of every team member in every story battle (best team).

Battle, team level	Pokémon	Held item	Moves
Rival (Cherrygrove), level 5	Totodile	None	Scratch, Leer
	Geodude	None	Tackle, Defense Curl
Falkner, level 11	Totodile	Mystic Water	Water Gun†, Scratch, Leer, Rage†
	Geodude	Hard Stone	Rock Throw†, Tackle, Defense Curl
Proton (Slowpoke Well), level 11	Totodile	Sitrus Berry	Rock Tomb†, Water Gun, Scratch, Rage
	Geodude	Hard Stone	Rock Throw, Tackle, Defense Curl
Bugsy, level 15	Totodile	Sitrus Berry	Rock Tomb, Bite†, Water Gun, Scratch
	Growlithe	Sitrus Berry	Ember, Bite, Leer, Roar
	Geodude	Hard Stone	Rock Throw, Tackle, Defense Curl
Rival (Azalea), level 15	Totodile	Sitrus Berry	Bite, Rock Tomb, Water Gun, Scratch
	Growlithe	Sitrus Berry	Bite, Ember, Leer, Roar
	Geodude	Hard Stone	Rock Throw, Tackle, Defense Curl
Whitney, level 18	Croconaw	Sitrus Berry	Bite, Blizzard†, Water Gun, Scratch
	Arcanine	Charcoal	Fire Blast†, Bite, Ember, Solar Beam†
	Geodude	Hard Stone	Rock Throw, Dig†, Fire Blast†, Tackle
	Heracross	Sitrus Berry	Focus Blast†, Aerial Ace†, Horn Attack, Tackle
Rival (Burned Tower), level 18	Croconaw	Sitrus Berry	Bite, Blizzard, Rock Smash†, Water Gun
	Arcanine	Charcoal	Fire Blast, Bite, Ember, Solar Beam
	Geodude	Sitrus Berry	Rock Throw, Fire Blast, Dig, Tackle
	Heracross	Sitrus Berry	Aerial Ace, Rock Smash†, Horn Attack, Focus Blast

Battle, team level	Pokémon	Held item	Moves
Morty, level 20	Croconaw	Sitrus Berry	Rock Smash, Bite, Blizzard, Water Gun
	Arcanine	Sitrus Berry	Fire Blast, Bite, Solar Beam, Ember
	Geodude	Sitrus Berry	Dig, Rock Throw, Fire Blast, Rock Smash†
	Heracross	Sitrus Berry	Rock Smash, Aerial Ace, Brick Break†, Horn Attack
	Umbreon	Sitrus Berry	Payback†, Hyper Beam†, Tackle, Helping Hand
Chuck, level 28	Croconaw	Sitrus Berry	Rock Smash, Surf†, Ice Fang†, Strength†
	Arcanine	Charcoal	Fire Blast, Solar Beam, Strength†, Ember
	Graveler	Sitrus Berry	Rock Smash, Strength†, Fire Blast, Rollout†
	Heracross	Sharp Beak	Rock Smash, Aerial Ace, Brick Break, Strength†
	Lapras	Mystic Water	Surf†, Thunder†, Blizzard†, Water Pulse
	Umbreon	Sitrus Berry	Hyper Beam, Payback, Tackle, Helping Hand
Jasmine, level 28	Croconaw	Wave Incense	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Solar Beam, Ember
	Graveler	Sitrus Berry	Strength, Rock Smash, Fire Blast, Rollout
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Aerial Ace
	Lapras	Mystic Water	Surf, Thunder, Blizzard, Water Pulse
	Umbreon	Sitrus Berry	Hyper Beam, Payback, Tackle, Helping Hand
Petrel (Rocket Hideout), level 28	Croconaw	Sitrus Berry	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Solar Beam, Ember
	Graveler	Sitrus Berry	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Aerial Ace
	Lapras	Sitrus Berry	Surf, Thunder, Blizzard, Water Pulse
	Umbreon	Sitrus Berry	Hyper Beam, Payback, Tackle, Helping Hand
Ariana (Rocket Hideout), level 28	Croconaw	Sitrus Berry	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Solar Beam, Ember
	Graveler	Sitrus Berry	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Aerial Ace
	Lapras	Mystic Water	Surf, Blizzard, Thunder, Rock Smash†
	Umbreon	Sitrus Berry	Iron Tail†, Payback, Hyper Beam, Tackle
Pryce, level 28	Croconaw	Sitrus Berry	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Solar Beam, Ember
	Graveler	Sitrus Berry	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Black Belt	Strength, Rock Smash, Brick Break, Shadow Claw†
	Lapras	Magnet	Surf, Rock Smash, Thunder, Blizzard
	Umbreon	Sitrus Berry	Payback, Iron Tail, Hyper Beam, Tackle
Petrel (Radio Tower), level 28	Croconaw	Wave Incense	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Ember, Solar Beam
	Graveler	Sitrus Berry	Strength, Rock Smash, Fire Blast, Rollout
	Heracross	Sitrus Berry	Strength, Rock Smash, Shadow Claw, Brick Break
	Lapras	Mystic Water	Surf, Rock Smash, Blizzard, Thunder
	Umbreon	Sitrus Berry	Hyper Beam, Payback, Iron Tail, Tackle
Rival (Goldenrod Underground), level 28	Croconaw	Wave Incense	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Fire Blast, Rock Smash†, Ember
	Graveler	Sitrus Berry	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Shadow Claw, Brick Break
	Lapras	Sitrus Berry	Surf, Rock Smash, Blizzard, Thunder

Battle, team level	Pokémon	Held item	Moves
	Umbreon	Sitrus Berry	Payback, Iron Tail, Hyper Beam, Tackle
Proton (Radio Tower), level 28	Croconaw	Sitrus Berry	Surf, Strength, Rock Smash, Ice Fang
	Arcanine	Charcoal	Strength, Rock Smash, Fire Blast, Ember
	Graveler	Hard Stone	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Shadow Claw
	Lapras	Sitrus Berry	Surf, Rock Smash, Blizzard, Thunder
	Umbreon	Sitrus Berry	Hyper Beam, Payback, Iron Tail, Tackle
Ariana (Radio Tower), level 28	Croconaw	Sitrus Berry	Surf, Strength, Rock Smash, Blizzard†
	Arcanine	Charcoal	Strength, Rock Smash, Fire Blast, Ember
	Graveler	Hard Stone	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Shadow Claw
	Lapras	Sitrus Berry	Surf, Rock Smash, Blizzard, Thunder
	Umbreon	Sitrus Berry	Iron Tail, Payback, Hyper Beam, Tackle
Archer (Radio Tower), level 28	Croconaw	Mystic Water	Surf, Strength, Rock Smash, Blizzard
	Arcanine	Black Belt	Strength, Rock Smash, Fire Blast, Solar Beam†
	Graveler	Hard Stone	Strength, Rock Smash, Rollout, Fire Blast
	Heracross	Sitrus Berry	Strength, Rock Smash, Brick Break, Shadow Claw
	Lapras	Sitrus Berry	Surf, Rock Smash, Blizzard, Thunder
	Umbreon	Sitrus Berry	Payback, Hyper Beam, Iron Tail, Tackle
Clair, level 38	Feraligatr	Never-Melt Ice	Crunch†, Blizzard, Waterfall†, Surf
	Arcanine	Sitrus Berry	Fire Blast, Thunder Fang†, Strength, Fire Fang†
	Graveler	Soft Sand	Earthquake†, Rock Blast†, Fire Blast, Rock Smash
	Heracross	Sitrus Berry	Close Combat†, Shadow Claw, Aerial Ace†, Strength
	Lapras	Sitrus Berry	Ice Beam†, Thunder, Waterfall†, Surf
	Umbreon	BlackGlasses	Feint Attack†, Iron Tail, Hyper Beam, Payback
Rival (Victory Road), level 38	Feraligatr	Sitrus Berry	Waterfall, Earthquake†, Blizzard, Rock Smash†
	Arcanine	Sitrus Berry	Fire Blast, Bite†, Rock Smash†, Fire Fang
	Graveler	Sitrus Berry	Earthquake, Rock Blast, Fire Blast, Rock Smash
	Heracross	Sitrus Berry	Close Combat, Shadow Claw, Aerial Ace, Strength
	Lapras	Sitrus Berry	Ice Beam, Waterfall, Rock Smash†, Surf
	Umbreon	Sitrus Berry	Dark Pulse†, Iron Tail, Feint Attack, Payback
Will, level 54	Feraligatr	Sitrus Berry	Waterfall, Ice Fang†, Crunch†, Earthquake
	Arcanine	Sitrus Berry	Fire Blast, Thunder Fang†, Dragon Pulse†, Fire Fang
	Graveler	Sitrus Berry	Stone Edge†, Earthquake, Rollout†, Fire Blast
	Heracross	Sitrus Berry	Close Combat, Shadow Claw, Aerial Ace, Strength
	Lapras	Sitrus Berry	Ice Beam, Waterfall, Thunder†, Surf
	Umbreon	Sitrus Berry	Dark Pulse, Iron Tail, Feint Attack, Hyper Beam†
Koga, level 54	Feraligatr	Sitrus Berry	Ice Fang, Crunch, Waterfall, Earthquake
	Arcanine	Sitrus Berry	Fire Blast, Thunder Fang, Fire Fang, Dragon Pulse
	Graveler	Sitrus Berry	Fire Blast, Stone Edge, Rollout, Earthquake
	Heracross	Sitrus Berry	Close Combat, Strength, Aerial Ace, Shadow Claw
	Lapras	Mystic Water	Surf, Waterfall, Thunder, Ice Beam
	Umbreon	BlackGlasses	Iron Tail, Feint Attack, Hyper Beam, Dark Pulse
Bruno, level 54	Feraligatr	Sitrus Berry	Ice Fang, Earthquake, Crunch, Waterfall
	Arcanine	Charcoal	Fire Blast, Thunder Fang, Fire Fang, Dragon Pulse
	Graveler	Sitrus Berry	Fire Blast, Stone Edge, Rollout, Earthquake

Battle, team level	Pokémon	Held item	Moves
	Heracross	Sitrus Berry	Close Combat, Strength, Aerial Ace, Shadow Claw
	Lapras	Mystic Water	Ice Beam, Surf, Waterfall, Thunder
	Umbreon	Sitrus Berry	Iron Tail, Dark Pulse, Feint Attack, Hyper Beam
Karen, level 54	Feraligatr	Sitrus Berry	Ice Fang, Crunch, Earthquake, Waterfall
	Arcanine	Sitrus Berry	Fire Blast, Thunder Fang, Fire Fang, Dragon Pulse
	Graveler	Sitrus Berry	Fire Blast, Stone Edge, Rollout, Earthquake
	Heracross	Sitrus Berry	Close Combat, Strength, Aerial Ace, Shadow Claw
	Lapras	Sitrus Berry	Surf, Waterfall, Thunder, Ice Beam
	Umbreon	Sitrus Berry	Iron Tail, Feint Attack, Hyper Beam, Dark Pulse
Lance, level 54	Feraligatr	Sitrus Berry	Ice Fang, Crunch, Earthquake, Waterfall
	Arcanine	Sitrus Berry	Fire Blast, Thunder Fang, Fire Fang, Dragon Pulse
	Graveler	Hard Stone	Fire Blast, Stone Edge, Rollout, Earthquake
	Heracross	Black Belt	Close Combat, Strength, Aerial Ace, Shadow Claw
	Lapras	Sitrus Berry	Ice Beam, Surf, Waterfall, Thunder
	Umbreon	BlackGlasses	Iron Tail, Dark Pulse, Feint Attack, Hyper Beam
Brock, level 54	Feraligatr	Sitrus Berry	Surf†, Earthquake, Ice Beam†, Focus Blast†
	Arcanine	Dragon Fang	Dragon Pulse, Fire Blast, Solar Beam†, Thunder Fang
	Graveler	Soft Sand	Earthquake, Stone Edge, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Aerial Ace, Shadow Claw, Strength
	Lapras	Sitrus Berry	Surf, Ice Beam, Thunder, Waterfall
	Umbreon	BlackGlasses	Dark Pulse, Dig†, Psychic†, Feint Attack
Misty, level 54	Feraligatr	Sitrus Berry	Earthquake, Ice Beam, Crunch†, Surf
	Arcanine	Sitrus Berry	Fire Blast, Solar Beam, Thunder Fang, Fire Fang†
	Graveler	Soft Sand	Earthquake, Stone Edge, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Rock Slide†, Shadow Claw, Aerial Ace
	Lapras	Magnet	Ice Beam, Thunder, Hydro Pump†, Waterfall
	Umbreon	BlackGlasses	Dark Pulse, Psychic, Dig, Feint Attack
Lt. Surge, level 54	Feraligatr	Sitrus Berry	Earthquake, Ice Beam, Crunch, Waterfall†
	Arcanine	Charcoal	Fire Blast, Fire Fang, Bite†, Solar Beam
	Graveler	Sitrus Berry	Earthquake, Stone Edge, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Rock Slide, Shadow Claw, Aerial Ace
	Lapras	Mystic Water	Ice Beam, Hydro Pump, Waterfall, Thunder
	Umbreon	BlackGlasses	Dark Pulse, Dig, Psychic, Feint Attack
Erika, level 54	Feraligatr	Sitrus Berry	Ice Beam, Waterfall, Crunch, Earthquake
	Arcanine	Charcoal	Fire Blast, Bite, Dig†, Fire Fang
	Graveler	Sitrus Berry	Stone Edge, Earthquake, Fire Blast, Rollout
	Heracross	Black Belt	Close Combat, Rock Slide, Shadow Claw, Aerial Ace
	Lapras	Leftovers	Ice Beam, Waterfall, Hydro Pump, Thunder
	Umbreon	BlackGlasses	Dark Pulse, Psychic, Feint Attack, Dig
Janine, level 54	Feraligatr	Wave Incense	Waterfall, Earthquake, Ice Beam, Crunch
	Arcanine	Leftovers	Fire Blast, Fire Fang, Bite, Dig
	Graveler	Hard Stone	Stone Edge, Earthquake, Rollout, Fire Blast
	Heracross	Rock Incense	Rock Slide, Close Combat, Shadow Claw, Aerial Ace
	Lapras	Mystic Water	Ice Beam, Waterfall, Hydro Pump, Rock Smash†
	Umbreon	TwistedSpoon	Psychic, Feint Attack, Dig, Dark Pulse
Sabrina, level 54	Feraligatr	Sitrus Berry	Waterfall, Crunch, Ice Beam, Earthquake
	Arcanine	BlackGlasses	Bite, Fire Blast, Thunder Fang†, Dig
	Graveler	Soft Sand	Earthquake, Stone Edge, Fire Blast, Rollout

Battle, team level	Pokémon	Held item	Moves
	Heracross	Leftovers	Night Slash†, Close Combat, Rock Slide, Shadow Claw
	Lapras	Mystic Water	Waterfall, Ice Beam, Surf†, Thunder†
	Umbreon	Sitrus Berry	Feint Attack, Dark Pulse, Psychic, Dig
Blaine, level 54	Feraligatr	Sitrus Berry	Waterfall, Ice Beam, Earthquake, Surf†
	Arcanine	Soft Sand	Fire Blast, Dig, Thunder Fang, Bite
	Graveler	Sitrus Berry	Earthquake, Stone Edge, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Night Slash, Rock Slide, Shadow Claw
	Lapras	Sitrus Berry	Surf, Ice Beam, Waterfall, Thunder
	Umbreon	BlackGlasses	Dark Pulse, Feint Attack, Psychic, Dig
Rival (Mt. Moon), level 54	Feraligatr	Sitrus Berry	Waterfall, Earthquake, Ice Beam, Rock Smash†
	Arcanine	Sitrus Berry	Overheat†, Bite, Thunder Fang, Fire Fang†
	Graveler	Sitrus Berry	Earthquake, Stone Edge, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Night Slash, Rock Slide, Shadow Claw
	Lapras	Sitrus Berry	Ice Beam, Waterfall, Surf, Rock Smash†
	Umbreon	Sitrus Berry	Dark Pulse, Feint Attack, Dig, Psychic
Blue, level 57	Feraligatr	Sitrus Berry	Waterfall, Ice Beam, Thrash†, Surf†
	Arcanine	Sitrus Berry	Overheat, Thunder Fang, Solar Beam†, Fire Fang
	Graveler	Sitrus Berry	Stone Edge, Earthquake, Fire Blast, Rollout
	Heracross	Leftovers	Close Combat, Megahorn†, Rock Slide, Night Slash
	Lapras	Sitrus Berry	Ice Beam, Hydro Pump†, Thunder†, Surf
	Umbreon	BlackGlasses	Dark Pulse, Psychic, Feint Attack, Dig

Note: † marks a move learned by level-up, taught by TM or HM, or relearned since the previous battle. Before Clair, a move that is dropped cannot be learned again (the Move Relearner is in Blackthorn City); HM moves cannot be dropped before then. Source: Author's own calculations.